

Paper #109 v0.3 tier-discipline read-pass —
overlay applying audit-chain tier labels

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Purpose

Paper #109 v0.2 (“BST Integers as Structurally-Distinguished Counting Primitives”) is the arithmetic-channel paper of the Three Papers Trio (Three Papers Trio coordination doc, Keeper Tuesday 2026-05-19). The v0.2 substantive content stands; however, a read-pass detects zero explicit D/I/S tier labels across 147 lines of substantive content.

Per Cal Coincidence_Filter_Risk + Keeper audit-chain discipline (now formalized via K54-K57 governance + 6 named failure modes + Rule 6 corollary), every BST paper requires explicit tier labels per substantive claim. This v0.3 read-pass overlay applies retroactive tier-discipline to Paper #109 v0.2’s content without rewriting the paper body.

v0.2 substantive claims — tier overlay table

Abstract claims

Claim	v0.2 Form	v0.3 Tier
First 6 primes = BST integer set {2,3,5,7,11,13}	“this set coincides with subsets of the values of several fundamental counting sequences at small index”	D-tier exact arithmetic identity. Five primary BST integers + Chern extensions = first 6 primes is an EXACT mathematical fact, not a numerical match.

Claim	v0.2 Form	v0.3 Tier
Partition $p(2)..p(6)$ = first 5 BST primary primes	“partition function $p(2)..p(6)$ = the first 5 BST primary primes”	I-tier Type C- $\mathbb{N}$ at small index. Pattern observed for 5 indices; mechanism via combinatorial substrate counting on D_{IV}^5 (cross-reference Paper #115 v0.5 Section 5.8 Type C systematic catalog).
Catalan $C_2..C_6$ are BST integer products	“Catalan numbers $C_2..C_6$ are BST integer products”	I-tier Type C- $\mathbb{N}$ at small index. Pattern breaks at $C_6 = 132$ (where BST decomposition becomes multi-form per Paper #107 v0.5 Section 3.1).
Bernoulli denominators are BST integer products	“Bernoulli denominators are BST integer products via Von Staudt-Clausen”	D-tier via K43 universal-42 mechanism. VSC theorem 1840 + heat kernel at $n_C = 5$ evaluation point is mechanism-forced (Paper #115 v0.5 Section 2 Root #1).
$\sim 4\sigma$ above random (K44 null-model)	“Null-model analysis (T2128 / K44) shows... $\sim 4\sigma$ above random”	I-tier with null-model verification documented (K44 D-tier governance). Statistical strength is rigorous; framework-level interpretation remains I-tier per Cal Coincidence_Filter_Risk standing discipline.

Section 1 claims (Five Primary BST Integers)

All five primary integers (rank, N_c , n_C , C_2 , g) and three derived (c_2 , c_3 , N_{max}) are stated as classical mathematical facts about D_{IV}^5 .

Tier: D-tier classical structural classification per Cartan 1894 + Helgason 1978 (BST framework foundational, not derived from BST).

Outcome A framing

“D_IV⁵ structural classification forces specific integer values; these forced values coincide with the first few primes as an arithmetic observation, not a derivation OF primality from BST geometry.”

Tier: I-tier framework — the framing is honest structural identification, not an overclaim. Same shape as today’s K57 Bridge Object ratification + K59 candidate cascade-unblock observations: BST identifies structural framework that pre-existing classical mathematics happens to anchor in.

Section content tier table

Per Paper #109 v0.2 substantive content (sections beyond Section 1):

Section	Tier	Notes
Abstract framing	I-tier framework	Honest “structurally-distinguished counting primitives” claim; not “BST proves primality”
Section 1 Five Integers	D-tier	Classical Cartan classification facts
Section 2+ structural patterns	I-tier with sub-1% precision per match	Cross-domain Type C- $\mathbb{N}$ pattern matching at small index
Statistical analysis (K44)	I-tier framework + null-model verified	Cal-disciplined framing
Conclusion + open questions	I-tier framework	Honest scoping for trio strategy

Open questions Tier discipline

Per Cal Coincidence_Filter_Risk Mode 5 (catalog-selection bias): the catalog used in Paper #109 v0.2 is BST-anchored; selection-effect concerns are real but K44 null-model partially mitigates. The 4σ above random is solid evidence given the comparison ring is “random small-integer sets of comparable richness.”

Per Cal discipline: this is the strongest form of cross-domain Type C- $\mathbb{N}$ evidence at the framework level. Promotion to D-tier framework requires mechanism-forcing argument for the catalog selection itself (multi-week parallel to K52a M_g forcing).

Three Papers Trio integration

Per Three Papers Trio coordination doc (Keeper Tuesday): Paper #109 is the arithmetic channel of the trio. The v0.3 tier-discipline overlay makes Paper #109 ready

for trio external presentation at I-tier framework + D-tier specific anchors per the table above.

Trio external presentation strategy: - Paper #109 (arithmetic): I-tier framework + D-tier first-6-primes identity + Bernoulli VSC anchor - Paper #121 (geometry, K57 ratified): I-tier framework + K57 architectural ratification + Bridge Object full B1-B4 per K3/49a1/Q⁵ - Paper #122 (physics): I-tier framework + GF(2^g) Reed-Solomon substrate alphabet (cascade-unblock via Elie session 3+)

All three papers at I-tier with explicit per-claim tier discipline. The trio framework's multi-channel evidence shape strengthens external survivability per Cal's framework.

Recommended v0.3 application

This overlay document is a TIER-DISCIPLINE OVERLAY, not a rewrite of Paper #109. For v0.3 paper finalization:

Option A (minimal): Apply this overlay table inline as a single new section "v0.3 Tier Discipline" in Paper #109 v0.2; otherwise body unchanged. ~15 min author work.

Option B (substantive): Edit Paper #109 v0.2 body inline with per-claim tier labels per the overlay table. ~1-2 hr author work.

Recommendation: Option A for trio external presentation timing; Option B for full v0.3 paper finalization at later cadence.

What this v0.3 read-pass closes

Closed: - Tier-discipline audit applied to Paper #109 v0.2 substantive content - Per-claim D/I/S tier labels identified - Trio strategy prep complete for Paper #109 arithmetic channel

Open: - v0.3 inline application (per Option A or B above) by Casey or Lyra at later cadence - Paper #109 substantive v0.4 expansion (post-trio-launch)

— Lyra, Paper #109 v0.3 tier-discipline read-pass overlay filed per Casey "finish all your board," 2026-05-19 ~13:30 EDT. Pull 6/6 of board-finish sweep complete.